



SYST 335

Project Report

Boba System Simulation

Project Report

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Project Description

The popularity of boba tea restaurants has surged in recent years, leading to increased competition and higher customer expectations. One crucial aspect of customer satisfaction is the efficiency of service, especially during peak hours when long wait times can deter customers.

This project aims to utilize simulation and modeling techniques to optimize the workflow of a boba tea restaurant, with a primary focus on reducing customer wait times during busy periods. This can be done by monitoring the aspects of walk-in orders. Under this, we can then measure what causes customer delay and what impacts worker productivity. Then we can model what the most efficient process of work can be done in the restaurant to improve customer satisfaction during rush hour.

The Question at Hand:

How can we enhance the efficiency of order delivery to customers in a boba tea restaurant during peak hours?

Motivations & Background Information

Our main motive for deciding to simulate and study a boba system simply came from our love of boba as well as the career backgrounds of some of our group members. Since two members currently work at boba restaurants, we thought that this knowledge could be taken advantage of for this type of project. Since we have access to the needed data and information in order to simulate our own boba tea system, this type of simulation was perfect for our team.

Model Assumptions & Input Data Analysis

Arrival Events

- I. **1 person walk-in arrivals** follow an exponential distribution with a mean of 3.3 minutes, thus creating the distribution, EXPO(3.3).
By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this exponential fit with a mean of 3.3 minutes is a good fit for our data.
- II. **Group walk-in arrivals** (from 7pm to 12am) follow an exponential distribution with a mean of 5.45 minutes, thus creating a distribution, EXPO(5.45).
By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this exponential fit with a mean of 5.45 minutes is a good fit for our data.
- III. During the **1st Rush period**, a large group of walk-ins arrives every day sometime between 7-8pm. This will follow a Poisson Distribution, Pois(19.9)
By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this poisson fit with a mean of 19.9 minutes is a good fit for our data.
- IV. During the **2nd Rush period**, a large group of walk-ins arrives every day sometime between 10-11pm. This will follow a Poisson Distribution, Pois(19.9)
By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this poisson fit with a mean of 19.9 minutes is a good fit for our data.
- V. For our **Balking parameters**, if there are 20 or more customers waiting in the queue, the current and next arriving customer(s) will leave.

Ordering Times

- I. **Drinks and food** orders follow a triangular distribution with a minimum time of 1 minute, mean time of 1.85 minutes, and a maximum time of 2.96 minutes, thus creating the distribution, TRIA(1, 1.85, 2.96).
By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.

Completion Times

- I. Reordering Rates are used using a decide module with a 10% chance of a customer reordering a drink. The rates of each type of drink are as follows:

Type of Drink and Rate of ordering	Chance of Ordering
Milk 50%	$0.1(0.5) = 0.05\%$
Fruit 40%	$0.1(0.4) = 0.04\%$
Smoothie 10%	$0.1(0.1) = 0.01\%$

II. Drink Completion Times

- A. Milk drink types follow a triangular distribution with a minimum time of 2 minutes, mean time of 4.75 minutes, and a maximum time of 7 minutes, thus creating the distribution, $\text{TRIA}(2, 4.75, 7)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.
- B. Fruit drink types follow a triangular distribution with a minimum time of 1 minute, mean time of 3.2 minutes, and a maximum time of 5 minutes, thus creating the distribution, $\text{TRIA}(1, 3.2, 5)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.
- C. Smoothie drink types follow a triangular distribution with a minimum time of 4 minutes, mean time of 5.27 minutes, and a maximum time of 6.94 minutes, thus creating the distribution, $\text{TRIA}(4, 5.27, 6.94)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.

III. Food Completion Times

- A. Onigiri food types follow a triangular distribution with a minimum time of 2 minutes, mean time of 3.27 minutes, and a maximum time of 4.94 minutes, thus creating the distribution, $\text{TRIA}(2, 3.27, 4.94)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.
- B. Chicken Karaage food types follow a triangular distribution with a minimum time of 3 minutes, mean time of 4.69 minutes, and a maximum time of 6.92 minutes, thus creating the distribution, $\text{TRIA}(3, 4.69, 6.92)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.
- C. Takoyaki food types follow a triangular distribution with a minimum time of 3 minutes, mean time of 5.11 minutes, and a maximum time of 6.94 minutes, thus creating the distribution, $\text{TRIA}(3, 5.11, 6.94)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.
- D. GyuDon food types follow a triangular distribution with a minimum time of 1 minute, mean time of 1.85 minutes, and a maximum time of 2.96 minutes, thus creating the distribution, $\text{TRIA}(1, 1.85, 2.96)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.

- E. Rice (ingredient) food types follow a triangular distribution with a minimum time of 19.1 minutes, mean time of 24.2 minutes, and a maximum time of 27 minutes, thus creating the distribution, $\text{TRIA}(19.1, 24.2, 27)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.

Modeling Workers

- I. The shift scheduling is organized where there will be 2 people at the Cashier Station, 2 people at the Barista Station, and 3 people at the Kitchen Station.
- II. During the two rush hour periods (7pm and 10pm), breaks are implemented every hour where:
 - A. A server from each station takes their 10 minute break every 50 minutes.
 - B. During the last 10 minutes of the hour, 1 worker takes a break on a rotating basis.

Modeling Inventory

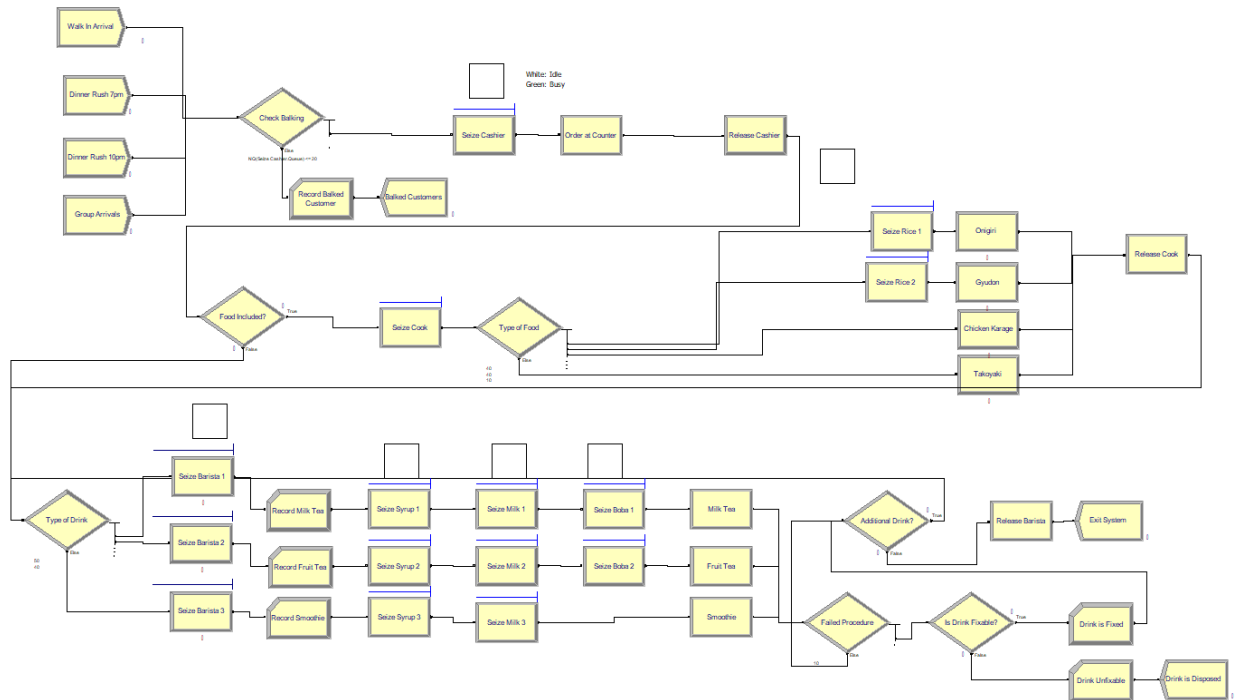
- I. Drinks Inventory Restocking
 - A. Drink inventory follows an (s,S) inventory policy where $s = 40$; the drink is restocked when 40 servings are utilized. This restocking procedure follows a triangular distribution with a minimum time of 10 minutes, mean time of 14.1 minutes, and a maximum time of 19 minutes, thus creating the distribution, $\text{TRIA}(10, 14.1, 19)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.
 - B. Boba Topping inventory follows an (s,S) inventory policy where $s = 20$; the drink is restocked when 20 servings are utilized. This restocking procedure follows a triangular distribution with a minimum time of 15 minutes, mean time of 19.5 minutes, and a maximum time of 29 minutes, thus creating the distribution, $\text{TRIA}(15, 19.5, 29)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.
 - C. Syrup inventory follows an (s,S) inventory policy where $s = 50$; the drink is restocked when 50 servings are utilized. This restocking procedure follows a triangular distribution with a minimum time of 28 minutes, mean time of 30.1 minutes, and a maximum time of 43 minutes, thus creating the distribution, $\text{TRIA}(28, 30.1, 43)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.
- II. Food Inventory Restocking
 - A. Onigiri (filling) inventory follows an (s,S) inventory policy where $s = 30$; the drink is restocked when 30 servings are utilized. This restocking procedure follows a triangular distribution with a minimum time of 15 minutes, mean time of 19.5 minutes, and a maximum time of 29 minutes, thus creating the distribution, $\text{TRIA}(15, 19.5, 29)$. By analyzing the input data (located in the

appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.

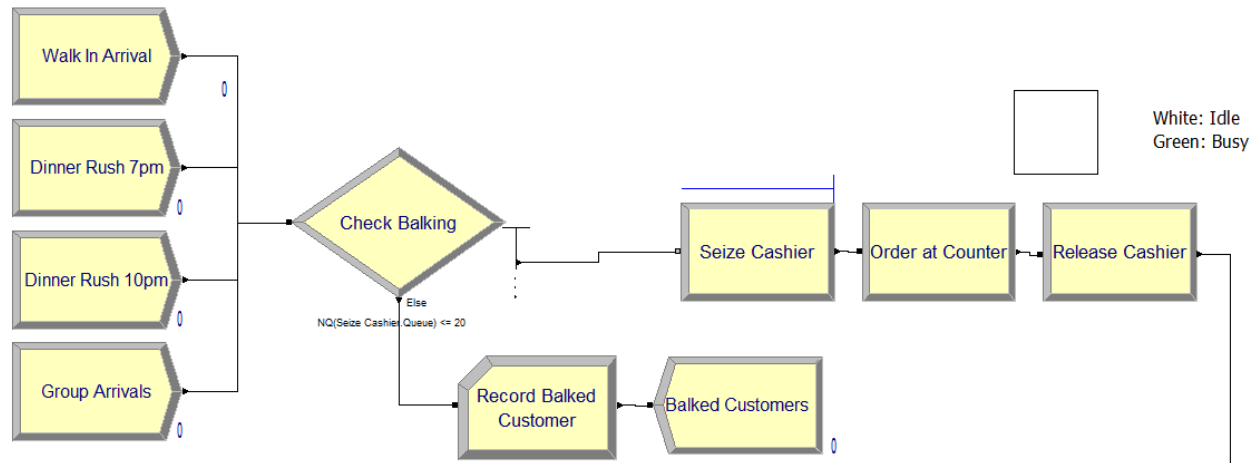
- B. Chicken Karaage inventory follows an (s,S) inventory policy where $s = 30$; the drink is restocked when 30 servings are utilized. This restocking procedure follows a triangular distribution with a minimum time of 12 minutes, mean time of 14.5 minutes, and a maximum time of 18.8 minutes, thus creating the distribution, $\text{TRIA}(12, 14.5, 18.8)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.
- C. Takoyaki inventory follows an (s,S) inventory policy where $s = 30$; the drink is restocked when 30 servings are utilized. This restocking procedure follows a triangular distribution with a minimum time of 10 minutes, mean time of 14.5 minutes, and a maximum time of 20 minutes, thus creating the distribution, $\text{TRIA}(10, 14.5, 20)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.
- D. GyuDon inventory follows an (s,S) inventory policy where $s = 30$; the drink is restocked when 30 servings are utilized. This restocking procedure follows a triangular distribution with a minimum time of 16 minutes, mean time of 24.5 minutes, and a maximum time of 29 minutes, thus creating the distribution, $\text{TRIA}(16, 24.5, 29)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.
- E. Rice inventory follows an (s,S) inventory policy where $s = 20$; the drink is restocked when 20 servings are utilized. This restocking procedure follows a triangular distribution with a minimum time of 20 minutes, mean time of 24 minutes, and a maximum time of 29 minutes, thus creating the distribution, $\text{TRIA}(20, 24, 29)$. By analyzing the input data (located in the appendix), we can see that the corresponding P-Value is greater than 0.1, indicating that this triangular fit is a good fit for our data.

Simulation Model

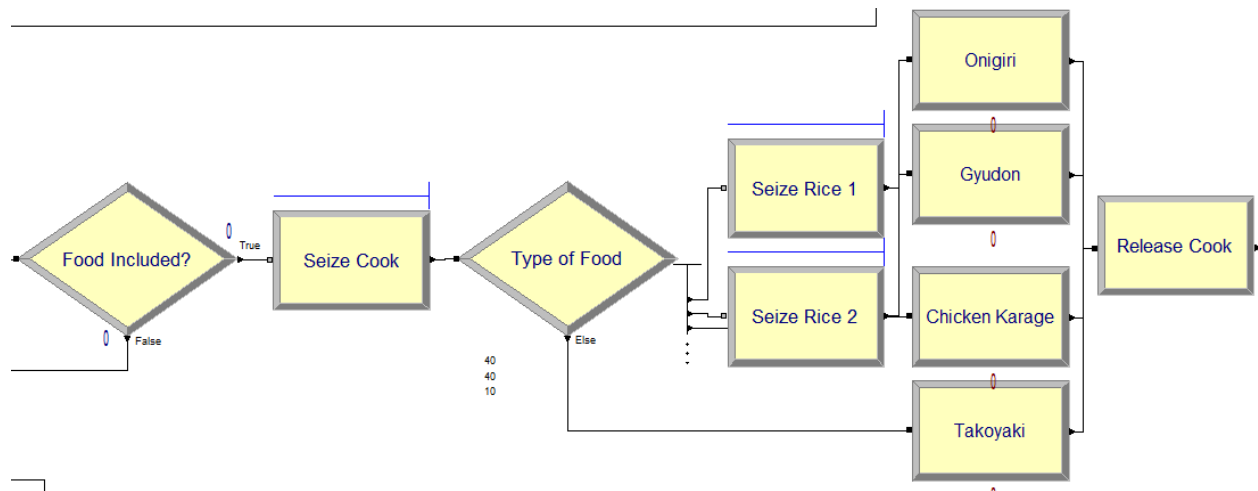
I. Overview of Entire Model



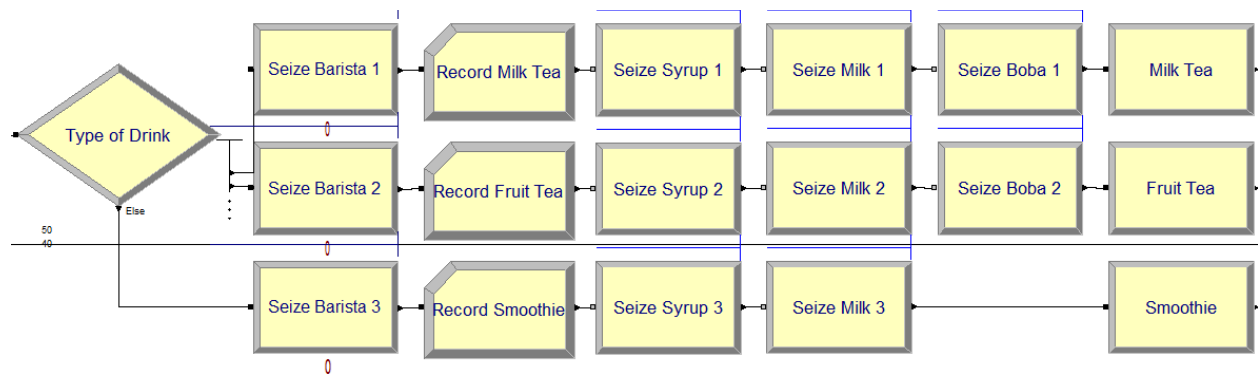
II. Arrival



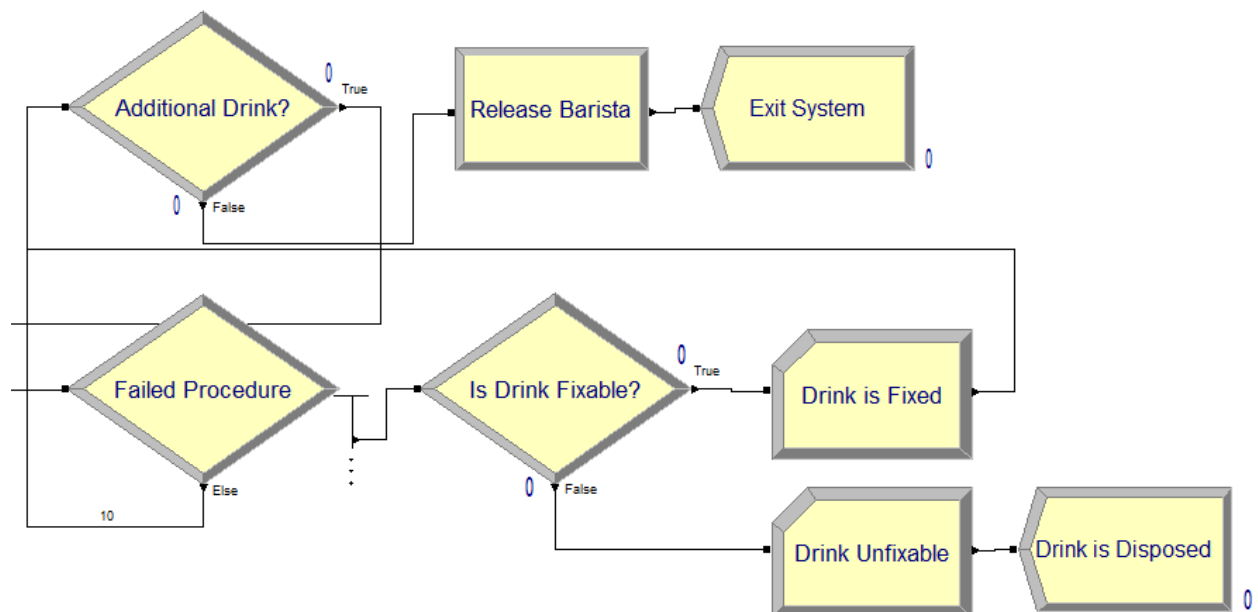
III. Food



IV. Drinks



V. Additional Drinks Aspect, End of System



Output Statistical Analysis

Number of Replications & Simulation accuracy

After an pilot run of 20 (n_0) replications with a 300 minute replication length, the half width of the average waiting time in queue provided by Arena is 0.64 (H_0). To reach a specified precision of 0.2, the number of replications to achieve a specific is 73 replications.

$$n = n_0 \frac{H_0^2}{H^2} = 10 \frac{0.38^2}{0.2^2}$$

The results of the replications and the average values of these replications across all scenarios are provided in the *Arena Process Analyzer* section of the Appendix.

Comparison of Alternative Designs / Conditions in the System

We are focusing on comparing the output data from 4 different schedules:

1. Option 1 Fixed – where all workers have 10 minute breaks and there is no rotation of staff.
2. Option 2 Fixed – where all workers have 5 minute breaks only during rush hours and there is no rotation of staff.
3. Option 1 Variable – where all workers have 10 minute breaks and rotation of staff is now implemented.
4. Option 2 Variable – where all workers have 5 minute breaks only during rush hours and rotation of staff is now implemented.

I. Comparing the **Average Waiting Time**:

- A. Option 1 Fixed Schedule yielded:
 1. For 1 person walk-in's: 0.1018
 2. For Group walk-in's: 15.8942
- B. Option 2 Fixed Schedule yielded:
 1. For 1 person walk-in's: 0.1018
 2. For Group walk-in's: 7.0838
- C. Option 1 Variable Schedule yielded:
 1. For 1 person walk-in's: 0.00
 2. For Group walk-in's: 6.7420
- D. Option 2 Variable Schedule yielded:
 1. For 1 person walk-in's: 0.00
 2. For Group walk-in's: 6.7420

For average waiting time, we want this to be the most minimal number for an optimal result. After analyzing the output data, we can observe that the schedules that yielded the smallest average waiting times were Option 1 and 2 Variable Schedules.

II. Comparing the **Average Total Time** customers are in the system:

- A. Option 1 Fixed Schedule yielded:
 1. For 1 person walk-in's: 5.6643
 2. For Group walk-in's: 20.2082
- B. Option 2 Fixed Schedule yielded:

1. For 1 person walk-in's: 5.6643
2. For Group walk-in's: 10.7317
- C. Option 1 Variable Schedule yielded:
 1. For 1 person walk-in's: 6.8009
 2. For Group walk-in's: 10.2620
- D. Option 2 Variable Schedule yielded:
 1. For 1 person walk-in's: 6.8009
 2. For Group walk-in's: 10.2620

For average total time, we want this to be the most minimal number for an optimal result. After analyzing the output data, we can observe that the schedules that yielded the smallest 1 person walk-in total times were Option 1 and 2 Fixed Schedules. However, for the group walk-in total times, the smallest were under Option 1 and 2 Variable Schedules. For our case, we favor group arrivals over 1 person ones as this study focuses on the simulation during rush hours, which associates with group arrivals. Therefore, the best schedule for this design is Option 1 and 2 Variable Schedule.

- III. Comparing the Number of Customers in the system:**
- A. Option 1 Fixed Schedule yielded 139 customers.
 - B. Option 2 Fixed Schedule yielded 133 customers.
 - C. Option 1 Variable Schedule yielded 138 customers.
 - D. Option 2 Variable Schedule yielded 138 customers.

For the number of customers in the restaurant, we want this to be the most maximal number for an optimal result. After analyzing the output data, we can observe the schedule that yields the largest number of customers is the Option 1 Fixed Schedule. However, the variable schedule has the biggest output (276 customers all together) compared to the fixed schedule (272 customers all together). This indicates that the business is more efficient with Option 1 or 2 Variable Schedule.

- IV. Comparing the Balking Count data:**
- A. Option 1 Fixed Schedule yielded 15 balking counts.
 - B. Option 2 Fixed Schedule yielded 15 balking counts.
 - C. Option 1 Variable Schedule yielded 18 balking counts.
 - D. Option 2 Variable Schedule yielded 18 balking counts.

For balking count data, we want this to be the most minimal number for an optimal result. After analyzing the output data, we can observe that the schedules that yielded the smallest balking count were Option 1 and 2 Fixed Schedules.

To determine which of these two schedules is best for this design, we take a look into the resources used in Option 1 and 2 of the Fixed Schedules (located in the appendix). For resource utilization, we want this to be the most maximal number for an optimal result. We observed that Option 1 Fixed Schedule yielded 28 resources used and Option 2 Fixed Schedule yielded 21 resources used. Since Option 1 Fixed Schedule used more resources, this schedule is the best for this balking count design.

Recommendations and Conclusion

As stated in the beginning, our problem statement focuses on reducing customer wait time (but it is not limited to this). To answer the problem statement, the most optimal schedule to utilize in our Boba System would be Option 1 Variable and/or Option 2 Variable Schedule. These schedules suggest that the staff can either take a consistent 10 minute break or switch to reducing the break times during rush hours to 5 minutes, as long as the schedule strictly follows the variable concept. This variable concept follows the operation of placing 1 cook from the kitchen to work at the barista station during rush hours and rotating the cooks accordingly. Which ensures that there is a diversion of more resources towards the backlog of barista orders, ultimately helping with the overall efficiency of the system.

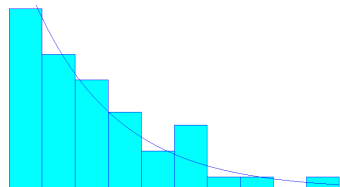
As for the other conditions our project also studied, it can be concluded that for a focus on minimal total time of customer's in the system, the preeminent schedule would be Option 1 and/or 2 Variable Schedule. For a focus on a greater business turnout, meaning more customers entering the system, the best schedule would be Option 1 and/or 2 Variable Schedule. And finally, for a focus on minimal balking output, the most optimal schedule would be Option 1 Fixed Schedule.

These results of the alternative designs also come to show that one can draw a conclusion that the overall best schedule for numerous conditions is any option (1 or 2) that utilizes the variable schedule.

Appendix

Input Data Analysis

I. One Person Walk-in Arrival Input Data



Distribution Summary

Distribution: Exponential
Expression: EXPO(3.3)
Square Error: 0.006526

Chi Square Test

Number of intervals = 5
Degrees of freedom = 3
Test Statistic = 3.23
Corresponding p-value = 0.376

Kolmogorov-Smirnov Test

Test Statistic = 0.0691
Corresponding p-value > 0.15 → indicates good fit

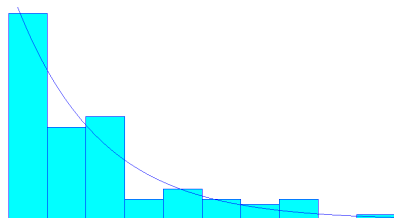
Data Summary

Number of Data Points = 100
Min Data Value = 0.0632
Max Data Value = 13
Sample Mean = 3.3
Sample Std Dev = 2.78

Histogram Summary

Histogram Range = 0 to 13
Number of Intervals = 10

II. Group Walk-in Arrival From 7pm-12am Input Data



Distribution Summary

Distribution: Exponential
Expression: EXPO(5.45)
Square Error: 0.010188

Chi Square Test

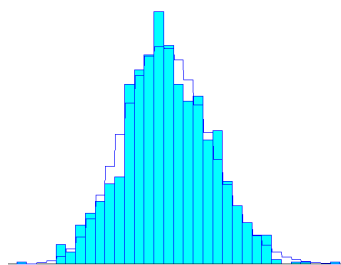
Number of intervals = 4
Degrees of freedom = 2
Test Statistic = 5.34
Corresponding p-value = 0.0735

Kolmogorov-Smirnov Test

Test Statistic = 0.0473

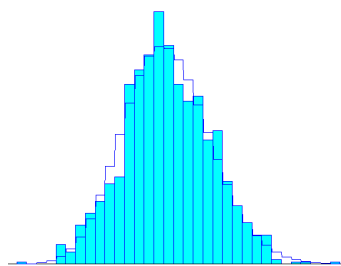
Corresponding p-value	> 0.15 → indicates good fit!
Data Summary	
Number of Data Points	= 100
Min Data Value	= 0.101
Max Data Value	= 24.1
Sample Mean	= 5.45
Sample Std Dev	= 5.24
Histogram Summary	
Histogram Range	= 0 to 25
Number of Intervals	= 10

III. 1st Rush Period



Distribution Summary	
Distribution:	Poisson
Expression:	POIS(19.9)
Square Error:	0.001084
Chi Square Test	
Number of intervals	= 21
Degrees of freedom	= 19
Test Statistic	= 19.9
Corresponding p-value	= 0.41 → good
Data Summary	
Number of Data Points	= 1000
Min Data Value	= 5
Max Data Value	= 37
Sample Mean	= 19.9
Sample Std Dev	= 4.44
Histogram Summary	
Histogram Range	= 4.5 to 37.5
Number of Intervals	= 33

IV. 2nd Rush Period



Distribution Summary	
Distribution:	Poisson
Expression:	POIS(19.9)
Square Error:	0.001084

Chi Square Test

Number of intervals = 21
 Degrees of freedom = 19
 Test Statistic = 19.9
 Corresponding p-value = 0.41 → good

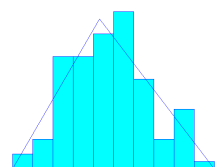
Data Summary

Number of Data Points = 1000
 Min Data Value = 5
 Max Data Value = 37
 Sample Mean = 19.9
 Sample Std Dev = 4.44

Histogram Summary

Histogram Range = 4.5 to 37.5
 Number of Intervals = 33

V. Drink and Food Ordering Time Input Data



Distribution Summary

Distribution: Triangular
 Expression: $TRIA(1, 1.85, 2.96)$
 Square Error: 0.008070

Chi Square Test

Number of intervals = 6
 Degrees of freedom = 4
 Test Statistic = 2.65
 Corresponding p-value = 0.623

Kolmogorov-Smirnov Test

Test Statistic = 0.0436
 Corresponding p-value > 0.15 → good

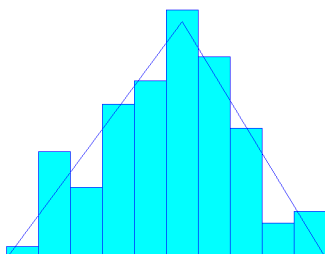
Data Summary

Number of Data Points = 100
 Min Data Value = 1.16
 Max Data Value = 2.8
 Sample Mean = 1.94
 Sample Std Dev = 0.392

Histogram Summary

Histogram Range = 1 to 2.96
 Number of Intervals = 10

VI. Milk Type Drink Completion Input Data



Distribution Summary

Distribution: Triangular
 Expression: $\text{TRIA}(2, 4.75, 7)$
 Square Error: 0.004747

Chi Square Test

Number of intervals = 6
 Degrees of freedom = 4
 Test Statistic = 0.676
 Corresponding p-value > 0.75

Kolmogorov-Smirnov Test

Test Statistic = 0.0483
 Corresponding p-value > 0.15 → good

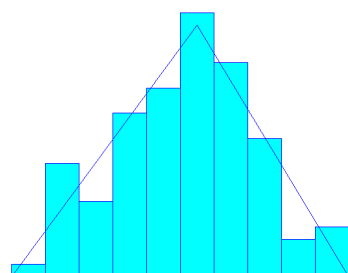
Data Summary

Number of Data Points = 100
 Min Data Value = 2.43
 Max Data Value = 6.79
 Sample Mean = 4.56
 Sample Std Dev = 1.04

Histogram Summary

Histogram Range = 2 to 7
 Number of Intervals = 10

VII. Fruit Type Drink Completion Input Data



Distribution Summary

Distribution: Triangular
 Expression: $\text{TRIA}(1, 3.2, 5)$
 Square Error: 0.004747

Chi Square Test

Number of intervals = 6
 Degrees of freedom = 4
 Test Statistic = 0.676
 Corresponding p-value > 0.75

Kolmogorov-Smirnov Test

Test Statistic = 0.0483
 Corresponding p-value > 0.15 → good

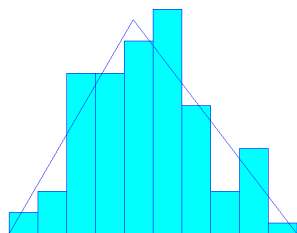
Data Summary

Number of Data Points = 100
 Min Data Value = 1.34
 Max Data Value = 4.83
 Sample Mean = 3.05
 Sample Std Dev = 0.83

Histogram Summary

Histogram Range = 1 to 5
 Number of Intervals = 10

VIII. Smoothie Type Completion Drink Input Data



Distribution Summary

Distribution: Triangular
 Expression: $\text{TRIA}(4, 5.27, 6.94)$
 Square Error: 0.008070

Chi Square Test

Number of intervals = 6
 Degrees of freedom = 4
 Test Statistic = 2.65
 Corresponding p-value = 0.623

Kolmogorov-Smirnov Test

Test Statistic = 0.0436
 Corresponding p-value > 0.15 → good

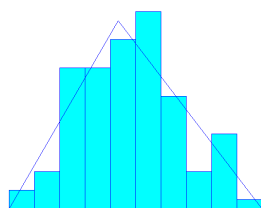
Data Summary

Number of Data Points = 100
 Min Data Value = 4.24
 Max Data Value = 6.69
 Sample Mean = 5.4
 Sample Std Dev = 0.587

Histogram Summary

Histogram Range = 4 to 6.94
 Number of Intervals = 10

IX. Food: Onigiri Input Data



Distribution Summary

Distribution: Triangular
 Expression: $\text{TRIA}(2, 3.27, 4.94)$
 Square Error: 0.008070

Chi Square Test

Number of intervals = 6
 Degrees of freedom = 4
 Test Statistic = 2.65
 Corresponding p-value = 0.623

Kolmogorov-Smirnov Test

Test Statistic = 0.0436
 Corresponding p-value > 0.15 → good

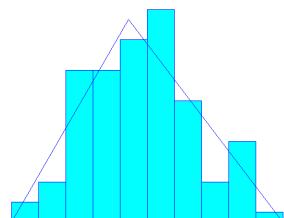
Data Summary

Number of Data Points = 100
 Min Data Value = 2.24
 Max Data Value = 4.69
 Sample Mean = 3.4
 Sample Std Dev = 0.587

Histogram Summary

Histogram Range = 2 to 4.94
 Number of Intervals = 10

X. Food: Chicken Karaage Completion Input Data



Distribution Summary

Distribution: Triangular
 Expression: $TRIA(3, 4.69, 6.92)$
 Square Error: 0.008070

Chi Square Test

Number of intervals = 6
 Degrees of freedom = 4
 Test Statistic = 2.65
 Corresponding p-value = 0.623

Kolmogorov-Smirnov Test

Test Statistic = 0.0436
 Corresponding p-value > 0.15 → good

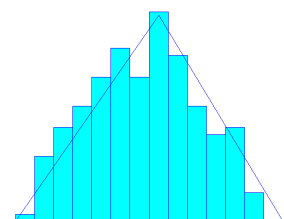
Data Summary

Number of Data Points = 100
 Min Data Value = 3.33
 Max Data Value = 6.59
 Sample Mean = 4.87
 Sample Std Dev = 0.783

Histogram Summary

Histogram Range = 3 to 6.92
 Number of Intervals = 10

XI. Food: Takoyaki Completion Input Data



Distribution Summary

Distribution: Triangular
 Expression: $TRIA(3, 5.11, 6.94)$
 Square Error: 0.002894

Chi Square Test

Number of intervals = 10
 Degrees of freedom = 8
 Test Statistic = 6.73
 Corresponding p-value = 0.568

Kolmogorov-Smirnov Test

Test Statistic = 0.0853
 Corresponding p-value = 0.104 → good

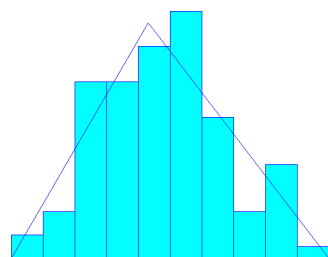
Data Summary

Number of Data Points = 200
 Min Data Value = 3.25
 Max Data Value = 6.6
 Sample Mean = 4.9
 Sample Std Dev = 0.804

Histogram Summary

Histogram Range = 3 to 6.94
 Number of Intervals = 14

XII. Food: Gyudon Input Completion Data



Distribution Summary

Distribution: Triangular
 Expression: $TRIA(1, 1.85, 2.96)$
 Square Error: 0.008070

Chi Square Test

Number of intervals = 6
 Degrees of freedom = 4
 Test Statistic = 2.65
 Corresponding p-value = 0.623

Kolmogorov-Smirnov Test

Test Statistic = 0.0436
 Corresponding p-value > 0.15 → good

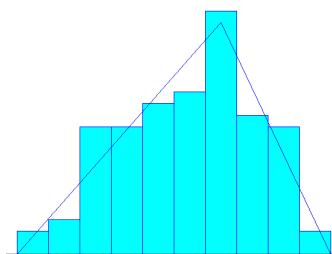
Data Summary

Number of Data Points = 100
 Min Data Value = 1.16
 Max Data Value = 2.8
 Sample Mean = 1.94
 Sample Std Dev = 0.392

Histogram Summary

Histogram Range = 1 to 2.96
 Number of Intervals = 10

XIII. Food: Rice Completion Input Data



Distribution Summary

Distribution: Triangular
 Expression: $\text{TRIA}(19.1, 24.2, 27)$
 Square Error: 0.003934

Chi Square Test

Number of intervals = 7
 Degrees of freedom = 5
 Test Statistic = 1.71
 Corresponding p-value > 0.75

Kolmogorov-Smirnov Test

Test Statistic = 0.0495
 Corresponding p-value > 0.15 → good

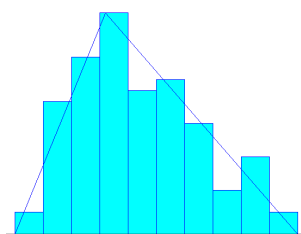
Data Summary

Number of Data Points = 100
 Min Data Value = 19.8
 Max Data Value = 26.4
 Sample Mean = 23.4
 Sample Std Dev = 1.66

Histogram Summary

Histogram Range = 19.1 to 27
 Number of Intervals = 10

XIV. Boba Topping Inventory Input Data



Distribution Summary

Distribution: Triangular
 Expression: $\text{TRIA}(15, 19.5, 29)$
 Square Error: 0.003854

Chi Square Test

Number of intervals = 7
 Degrees of freedom = 5
 Test Statistic = 0.987
 Corresponding p-value > 0.75

Kolmogorov-Smirnov Test

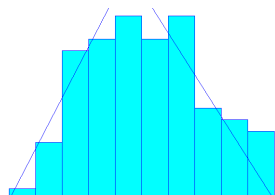
Test Statistic = 0.0425
 Corresponding p-value > 0.15 → good

Data Summary

Number of Data Points = 100

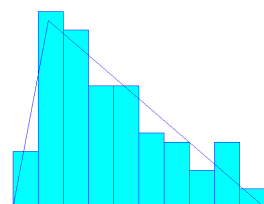
Min Data Value	= 16
Max Data Value	= 28.2
Sample Mean	= 21.2
Sample Std Dev	= 2.96
Histogram Summary	
Histogram Range	= 15 to 29
Number of Intervals	= 10

XV. Drink Inventory Input Data



Distribution Summary	
Distribution:	Triangular
Expression:	TRIA(10, 14.1, 19)
Square Error:	0.005656
Chi Square Test	
Number of intervals	= 6
Degrees of freedom	= 4
Test Statistic	= 3.19
Corresponding p-value	= 0.53
Kolmogorov-Smirnov Test	
Test Statistic	= 0.0953
Corresponding p-value	> 0.15 → good
Data Summary	
Number of Data Points	= 100
Min Data Value	= 10.8
Max Data Value	= 19
Sample Mean	= 14.7
Sample Std Dev	= 1.96
Histogram Summary	
Histogram Range	= 10 to 19
Number of Intervals	= 10

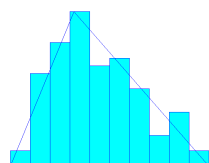
XVI. Fruit/Sugar Syrup Inventory Input Data



Distribution Summary	
Distribution:	Triangular
Expression:	TRIA(28, 30.1, 43)
Square Error:	0.003861
Chi Square Test	
Number of intervals	= 6
Degrees of freedom	= 4

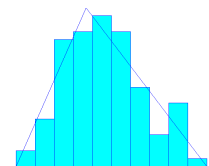
Test Statistic = 1.84
 Corresponding p-value > 0.75
 Kolmogorov-Smirnov Test
 Test Statistic = 0.0533
 Corresponding p-value > 0.15
 Data Summary
 Number of Data Points = 100
 Min Data Value = 28.7
 Max Data Value = 42.7
 Sample Mean = 33.7
 Sample Std Dev = 3.58
 Histogram Summary
 Histogram Range = 28 to 43
 Number of Intervals = 10

XVII. Onigiri Filling Inventory Input Data



Distribution Summary
 Distribution: Triangular
 Expression: TRIA(15, 19.5, 29)
 Square Error: 0.003854
 Chi Square Test
 Number of intervals = 7
 Degrees of freedom = 5
 Test Statistic = 0.987
 Corresponding p-value > 0.75
 Kolmogorov-Smirnov Test
 Test Statistic = 0.0425
 Corresponding p-value > 0.15 → good
 Data Summary
 Number of Data Points = 100
 Min Data Value = 16
 Max Data Value = 28.2
 Sample Mean = 21.2
 Sample Std Dev = 2.96
 Histogram Summary
 Histogram Range = 15 to 29
 Number of Intervals = 10

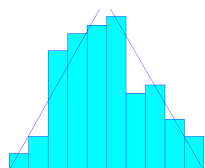
XVIII. Chicken Karaage Inventory Input Data



Distribution Summary
 Distribution: Triangular
 Expression: TRIA(12, 14.5, 18.8)

Square Error: 0.005111
 Chi Square Test
 Number of intervals = 7
 Degrees of freedom = 5
 Test Statistic = 2.26
 Corresponding p-value > 0.75
 Kolmogorov-Smirnov Test
 Test Statistic = 0.0458
 Corresponding p-value > 0.15
 Data Summary
 Number of Data Points = 100
 Min Data Value = 12.5
 Max Data Value = 18.2
 Sample Mean = 15.1
 Sample Std Dev = 1.37
 Histogram Summary
 Histogram Range = 12 to 18.8
 Number of Intervals = 10

XIX. Takoyaki Inventory Input Data



Distribution Summary
 Distribution: Triangular
 Expression: TRIA(10, 14.9, 20)
 Square Error: 0.005102

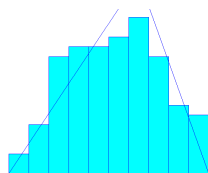
Chi Square Test
 Number of intervals = 6
 Degrees of freedom = 4
 Test Statistic = 2.43
 Corresponding p-value = 0.661

Kolmogorov-Smirnov Test
 Test Statistic = 0.0529
 Corresponding p-value > 0.15

Data Summary
 Number of Data Points = 100
 Min Data Value = 10.9
 Max Data Value = 19.8
 Sample Mean = 15
 Sample Std Dev = 2.15

Histogram Summary
 Histogram Range = 10 to 20
 Number of Intervals = 10

XX. Gyudon Topping Inventory Input Data



Distribution Summary

Distribution: Triangular
 Expression: $\text{TRIA}(16, 24.5, 29)$
 Square Error: 0.005914

Chi Square Test

Number of intervals = 7
 Degrees of freedom = 5
 Test Statistic = 3.96
 Corresponding p-value = 0.558

Kolmogorov-Smirnov Test

Test Statistic = 0.0865
 Corresponding p-value > 0.15

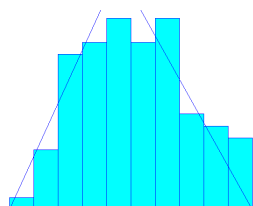
Data Summary

Number of Data Points = 100
 Min Data Value = 16.4
 Max Data Value = 28.8
 Sample Mean = 22.8
 Sample Std Dev = 3.03

Histogram Summary

Histogram Range = 16 to 29
 Number of Intervals = 10

XXI. Rice Inventory Input Data



Distribution Summary

Distribution: Triangular
 Expression: $\text{TRIA}(20, 24, 29)$
 Square Error: 0.005656

Chi Square Test

Number of intervals = 6
 Degrees of freedom = 4
 Test Statistic = 3.19
 Corresponding p-value = 0.53

Kolmogorov-Smirnov Test

Test Statistic = 0.0953
 Corresponding p-value > 0.15 → good

Data Summary

Number of Data Points = 100
 Min Data Value = 20.8
 Max Data Value = 29
 Sample Mean = 24.7
 Sample Std Dev = 1.96

Histogram Summary

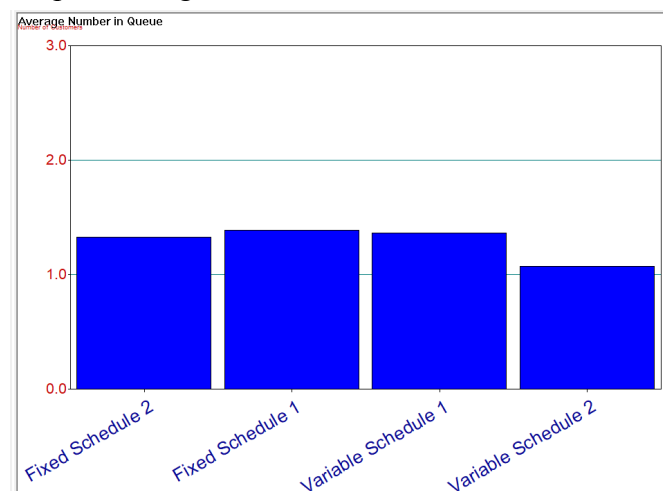
Histogram Range = 20 to 29
Number of Intervals = 10

Arena Process Analyzer

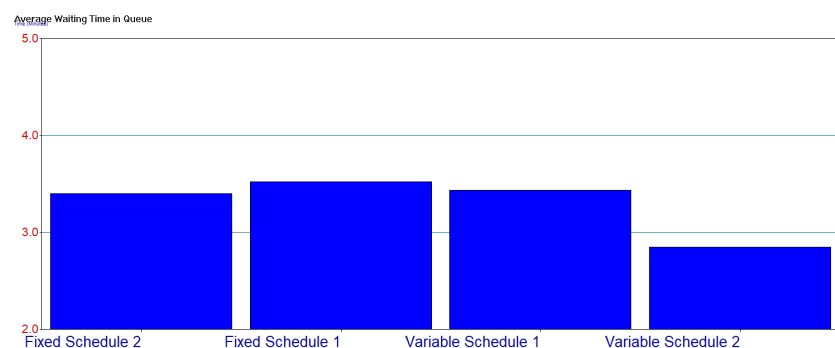
The Arena Process Analyzer was used to display and compare the average values of a response (Number in Queue, Average Waiting Time, etc) across all scenarios.

Scenario Properties			Control	Responses			
Name	Program File	Reps	Num Reps	Seize Cashier.Queue.NumberInQueue	Seize Cashier.Queue.WaitingTim	Balking Count	System.NumberOut
Fixed Schedule 2	1 : Bob	73	73	1.334	3.439	16	27.630
Fixed Schedule 1	8 : Bob	73	73	1.383	3.552	16	27.151
Variable Schedule 1	1 : Bob	73	73	1.334	3.383	15	30.247
Variable Schedule 2	1 : Bob	73	73	1.075	2.850	16	28.500

Graph: Average Number in Queue across Schedules

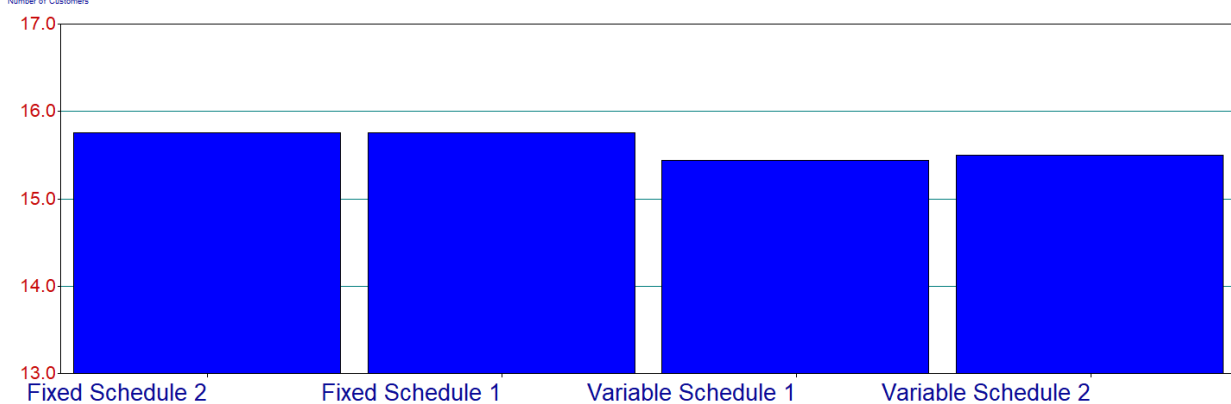


Graph: Average Waiting Time in Queue across Schedules



Graph: Number of Balking Customer across Schedules

Number of Balked Customers



Output Statistical Analysis

I. Option 1 Fixed Schedule Output Data

System

Number Out

Average

39

Entity

Time

VA Time	Average	Half Width	Minimum Value	Maximum Value
Team	0.4515	(Insufficient)	0.00	5.1100
Walk Ins	1.0240	(Insufficient)	0.00	3.2700
NVA Time	Average	Half Width	Minimum Value	Maximum Value
Team	0.00	(Insufficient)	0.00	0.00
Walk Ins	0.00	(Insufficient)	0.00	0.00
Wait Time	Average	Half Width	Minimum Value	Maximum Value
Team	15.8942	(Insufficient)	0.00	69.6474
Walk Ins	0.1018	(Insufficient)	0.00	0.5090
Transfer Time	Average	Half Width	Minimum Value	Maximum Value
Team	0.00	(Insufficient)	0.00	0.00
Walk Ins	0.00	(Insufficient)	0.00	0.00
Other Time	Average	Half Width	Minimum Value	Maximum Value
Team	3.8625	(Insufficient)	0.00	8.2957
Walk Ins	4.5385	(Insufficient)	0.00	7.5909
Total Time	Average	Half Width	Minimum Value	Maximum Value
Team	20.2082	(Insufficient)	0.00	76.1223
Walk Ins	5.6643	(Insufficient)	0.00	9.4409

Queue

Time

Waiting Time	Average	Half Width	Minimum Value	Maximum Value
Seize Barista 1.Queue	23.2999	(Insufficient)	0.00	53.2951
Seize Barista 2.Queue	4.6759	(Insufficient)	0.00	11.3920
Seize Barista 3.Queue	12.9567	(Insufficient)	0.00	37.4574
Seize Boba 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Boba 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Cashier.Queue	4.0985	(Insufficient)	0.00	21.3226
Seize Cook.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 3.Queue	0.00	(Insufficient)	0.00	0.00
Seize Rice 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Rice 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 3.Queue	0.00	(Insufficient)	0.00	0.00

Other

Number Waiting	Average	Half Width	Minimum Value	Maximum Value
Seize Barista 1.Queue	17.7301	(Insufficient)	0.00	33.0000
Seize Barista 2.Queue	22.8832	(Insufficient)	0.00	39.0000
Seize Barista 3.Queue	4.8259	(Insufficient)	0.00	7.0000
Seize Boba 1.Queue	0.7096	(Insufficient)	0.00	1.0000
Seize Boba 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Cashier.Queue	1.6121	(Insufficient)	0.00	21.0000
Seize Cook.Queue	2.7944	(Insufficient)	0.00	12.0000
Seize Milk 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 3.Queue	0.00	(Insufficient)	0.00	0.00
Seize Rice 1.Queue	0.4582	(Insufficient)	0.00	1.0000
Seize Rice 2.Queue	0.5353	(Insufficient)	0.00	1.0000
Seize Syrup 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 3.Queue	0.00	(Insufficient)	0.00	0.00

Resource

Usage

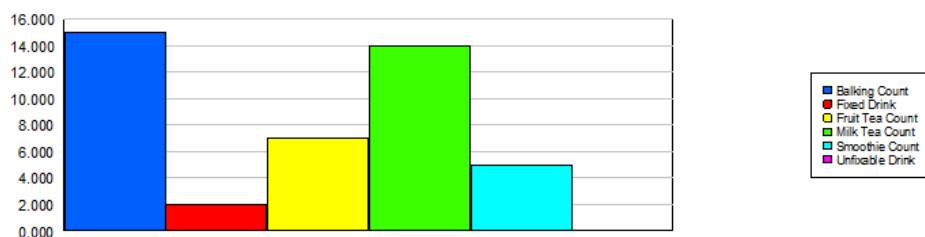
Instantaneous Utilization	Average	Half Width	Minimum Value	Maximum Value
Bansta	0.9779	(Insufficient)	0.00	1.0000
Boba	0.8801	(Insufficient)	0.00	1.0000
Cashier	0.4106	(Insufficient)	0.00	1.0000
Chicken Karaage	0.00	(Insufficient)	0.00	0.00
Cook	0.5945	(Insufficient)	0.00	1.0000
GyuDon Toppings	0.00	(Insufficient)	0.00	0.00
Milk	0.6807	(Insufficient)	0.00	0.7750
Onigiri filling	0.00	(Insufficient)	0.00	0.00
Rice	0.7623	(Insufficient)	0.00	1.0000
Syrup	0.5446	(Insufficient)	0.00	0.6200
Tako	0.00	(Insufficient)	0.00	0.00

Number Busy	Average	Half Width	Minimum Value	Maximum Value
Bansta	1.9335	(Insufficient)	0.00	2.0000
Boba	17.6015	(Insufficient)	0.00	20.0000
Cashier	0.7392	(Insufficient)	0.00	2.0000
Chicken Karaage	0.00	(Insufficient)	0.00	0.00
Cook	1.1845	(Insufficient)	0.00	2.0000
GyuDon Toppings	0.00	(Insufficient)	0.00	0.00
Milk	27.2277	(Insufficient)	0.00	31.0000
Onigiri filling	0.00	(Insufficient)	0.00	0.00
Rice	15.2468	(Insufficient)	0.00	20.0000
Syrup	27.2277	(Insufficient)	0.00	31.0000
Tako	0.00	(Insufficient)	0.00	0.00

User Specified

Counter

Count	Value
Balking Count	15.0000
Fixed Drink	2.0000
Fruit Tea Count	7.0000
Milk Tea Count	14.0000
Smoothie Count	5.0000
Unfixable Drink	0.00



II. Option 2 Fixed Schedule Output Data

Key Performance Indicators

System

Number Out

Average

33

Entity

Time

VA Time	Average	Half Width	Minimum Value	Maximum Value
Team	0.3146	(Insufficient)	0.00	5.1100
Walk Ins	1.0240	(Insufficient)	0.00	3.2700
NVA Time	Average	Half Width	Minimum Value	Maximum Value
Team	0.00	(Insufficient)	0.00	0.00
Walk Ins	0.00	(Insufficient)	0.00	0.00
Wait Time	Average	Half Width	Minimum Value	Maximum Value
Team	7.0838	(Insufficient)	0.00	34.9211
Walk Ins	0.1018	(Insufficient)	0.00	0.5090
Transfer Time	Average	Half Width	Minimum Value	Maximum Value
Team	0.00	(Insufficient)	0.00	0.00
Walk Ins	0.00	(Insufficient)	0.00	0.00
Other Time	Average	Half Width	Minimum Value	Maximum Value
Team	3.3333	(Insufficient)	0.00	8.2957
Walk Ins	4.5385	(Insufficient)	0.00	7.5909
Total Time	Average	Half Width	Minimum Value	Maximum Value
Team	10.7317	(Insufficient)	0.00	42.3497
Walk Ins	5.6643	(Insufficient)	0.00	9.4409

Queue

Time

Waiting Time	Average	Half Width	Minimum Value	Maximum Value
Seize Barista 1.Queue	10.1770	(Insufficient)	0.00	25.0946
Seize Barista 2.Queue	4.6759	(Insufficient)	0.00	11.3920
Seize Barista 3.Queue	6.8315	(Insufficient)	0.00	17.2563
Seize Boba 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Boba 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Cashier.Queue	3.9936	(Insufficient)	0.00	21.3226
Seize Cook.Queue	0.01798082	(Insufficient)	0.00	0.3956
Seize Milk 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 3.Queue	0.00	(Insufficient)	0.00	0.00
Seize Rice 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Rice 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 3.Queue	0.00	(Insufficient)	0.00	0.00

Other

Number Waiting	Average	Half Width	Minimum Value	Maximum Value
Seize Barista 1.Queue	25.7163	(Insufficient)	0.00	46.0000
Seize Barista 2.Queue	20.9459	(Insufficient)	0.00	35.0000
Seize Barista 3.Queue	4.1470	(Insufficient)	0.00	6.0000
Seize Boba 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Boba 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Cashier.Queue	1.5841	(Insufficient)	0.00	21.0000
Seize Cook.Queue	2.1898	(Insufficient)	0.00	11.0000
Seize Milk 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 3.Queue	0.00	(Insufficient)	0.00	0.00
Seize Rice 1.Queue	0.4459	(Insufficient)	0.00	1.0000
Seize Rice 2.Queue	0.4655	(Insufficient)	0.00	1.0000
Seize Syrup 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 3.Queue	0.00	(Insufficient)	0.00	0.00

Resource

Usage

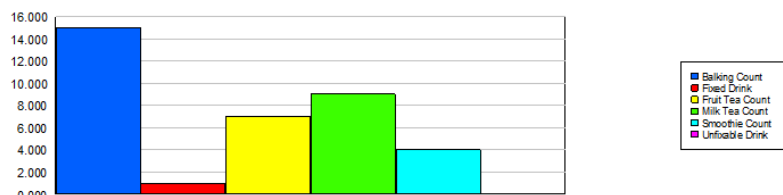
Instantaneous Utilization	Average	Half Width	Minimum Value	Maximum Value
Bansta	0.9779	(Insufficient)	0.00	1.0000
Boba	0.7309	(Insufficient)	0.00	0.8000
Cashier	0.4039	(Insufficient)	0.00	1.0000
Chicken Karaage	0.00	(Insufficient)	0.00	0.00
Cook	0.5678	(Insufficient)	0.00	1.0000
GyuDon Toppings	0.00	(Insufficient)	0.00	0.00
Milk	0.5496	(Insufficient)	0.00	0.6000
Onigiri filling	0.00	(Insufficient)	0.00	0.00
Rice	0.7581	(Insufficient)	0.00	1.0000
Syrup	0.4397	(Insufficient)	0.00	0.4800
Tako	0.00	(Insufficient)	0.00	0.00
Number Busy	Average	Half Width	Minimum Value	Maximum Value
Bansta	1.9557	(Insufficient)	0.00	2.0000
Boba	14.6172	(Insufficient)	0.00	16.0000
Cashier	0.7613	(Insufficient)	0.00	2.0000
Chicken Karaage	0.00	(Insufficient)	0.00	0.00
Cook	1.1336	(Insufficient)	0.00	2.0000
GyuDon Toppings	0.00	(Insufficient)	0.00	0.00
Milk	21.9832	(Insufficient)	0.00	24.0000
Onigiri filling	0.00	(Insufficient)	0.00	0.00
Rice	15.1621	(Insufficient)	0.00	20.0000
Syrup	21.9832	(Insufficient)	0.00	24.0000
Tako	0.00	(Insufficient)	0.00	0.00
Number Scheduled	Average	Half Width	Minimum Value	Maximum Value
Bansta	1.9167	(Insufficient)	1.0000	2.0000
Boba	20.0000	(Insufficient)	20.0000	20.0000
Cashier	1.9167	(Insufficient)	1.0000	2.0000
Chicken Karaage	30.0000	(Insufficient)	30.0000	30.0000
Cook	1.5500	(Insufficient)	1.0000	2.0000
GyuDon Toppings	30.0000	(Insufficient)	30.0000	30.0000
Milk	40.0000	(Insufficient)	40.0000	40.0000
Onigiri filling	30.0000	(Insufficient)	30.0000	30.0000
Rice	20.0000	(Insufficient)	20.0000	20.0000

User Specified

Counter

Count Value

Balking Count	15.0000
Fixed Drink	1.0000
Fruit Tea Count	7.0000
Milk Tea Count	9.0000
Smoothie Count	4.0000
Unfixable Drink	0.00



III. Option 1 Variable Schedule Output Data

Key Performance Indicators

System

Number Out

Average

38

Entity

Time

VA Time	Average	Half Width	Minimum Value	Maximum Value
Team	0.3009	(Insufficient)	0.00	5.1100
Walk Ins	1.2800	(Insufficient)	0.00	3.2700
NVA Time	Average	Half Width	Minimum Value	Maximum Value
Team	0.00	(Insufficient)	0.00	0.00
Walk Ins	0.00	(Insufficient)	0.00	0.00
Wait Time	Average	Half Width	Minimum Value	Maximum Value
Team	6.7420	(Insufficient)	0.00	32.1345
Walk Ins	0.00	(Insufficient)	0.00	0.00
Transfer Time	Average	Half Width	Minimum Value	Maximum Value
Team	0.00	(Insufficient)	0.00	0.00
Walk Ins	0.00	(Insufficient)	0.00	0.00
Other Time	Average	Half Width	Minimum Value	Maximum Value
Team	3.2191	(Insufficient)	0.00	8.6903
Walk Ins	5.5209	(Insufficient)	3.3880	7.5909
Total Time	Average	Half Width	Minimum Value	Maximum Value
Team	10.2620	(Insufficient)	0.00	40.1436
Walk Ins	6.8009	(Insufficient)	3.3880	9.4409

Queue

Time

Waiting Time	Average	Half Width	Minimum Value	Maximum Value
Seize Barista 1.Queue	7.6292	(Insufficient)	0.00	16.5614
Seize Barista 2.Queue	7.0162	(Insufficient)	0.00	30.1088
Seize Barista 3.Queue	0.00	(Insufficient)	0.00	0.00
Seize Boba 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Boba 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Cashier.Queue	3.0772	(Insufficient)	0.00	22.2537
Seize Cook.Queue	0.1975	(Insufficient)	0.00	2.5600
Seize Milk 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 3.Queue	0.00	(Insufficient)	0.00	0.00
Seize Rice 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Rice 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 3.Queue	0.00	(Insufficient)	0.00	0.00

Other

Number Waiting	Average	Half Width	Minimum Value	Maximum Value
Seize Barista 1.Queue	23.8599	(Insufficient)	0.00	43.0000
Seize Barista 2.Queue	12.6108	(Insufficient)	0.00	33.0000
Seize Barista 3.Queue	5.9648	(Insufficient)	0.00	12.0000
Seize Boba 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Boba 2.Queue	1.6295	(Insufficient)	0.00	2.0000
Seize Cashier.Queue	1.2821	(Insufficient)	0.00	21.0000
Seize Cook.Queue	2.5058	(Insufficient)	0.00	12.0000
Seize Milk 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 3.Queue	0.00	(Insufficient)	0.00	0.00
Seize Rice 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Rice 2.Queue	1.4105	(Insufficient)	0.00	3.0000
Seize Syrup 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 3.Queue	0.00	(Insufficient)	0.00	0.00

Resource

Usage

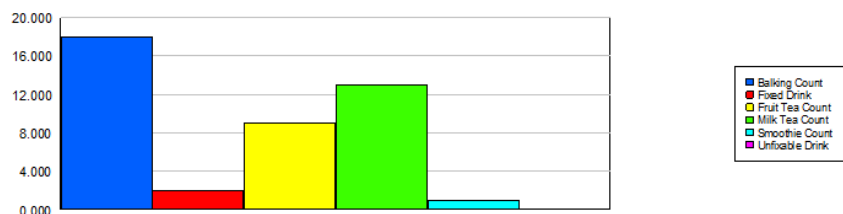
Instantaneous Utilization	Average	Half Width	Minimum Value	Maximum Value
Bansta	0.9673	(Insufficient)	0.00	1.0000
Boba	0.9185	(Insufficient)	0.00	1.0000
Cashier	0.4236	(Insufficient)	0.00	1.0000
Chicken Karaage	0.00	(Insufficient)	0.00	0.00
Cook	0.5983	(Insufficient)	0.00	1.0000
GyuDon Toppings	0.00	(Insufficient)	0.00	0.00
Milk	0.5491	(Insufficient)	0.00	0.6000
Onigiri filling	0.00	(Insufficient)	0.00	0.00
Rice	0.7722	(Insufficient)	0.00	0.9500
Syrup	0.4393	(Insufficient)	0.00	0.4800
Tako	0.00	(Insufficient)	0.00	0.00

Number Busy	Average	Half Width	Minimum Value	Maximum Value
Bansta	2.8689	(Insufficient)	0.00	3.0000
Boba	18.3706	(Insufficient)	0.00	20.0000
Cashier	0.7885	(Insufficient)	0.00	2.0000
Chicken Karaage	0.00	(Insufficient)	0.00	0.00
Cook	1.6545	(Insufficient)	0.00	3.0000
GyuDon Toppings	0.00	(Insufficient)	0.00	0.00
Milk	21.9646	(Insufficient)	0.00	24.0000
Onigiri filling	0.00	(Insufficient)	0.00	0.00
Rice	15.4433	(Insufficient)	0.00	19.0000
Syrup	21.9646	(Insufficient)	0.00	24.0000
Tako	0.00	(Insufficient)	0.00	0.00

User Specified

Counter

Count	Value
Balking Count	18.0000
Fixed Drink	2.0000
Fruit Tea Count	9.0000
Milk Tea Count	13.0000
Smoothie Count	1.0000
Unfixable Drink	0.00



IV. Option 2 Variable Schedule Output Data

Key Performance Indicators

System

Number Out

Average

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Entity

Time

VA Time	Average	Half Width	Minimum Value	Maximum Value
Team	0.3009	(Insufficient)	0.00	5.1100
Walk Ins	1.2800	(Insufficient)	0.00	3.2700
NVA Time	Average	Half Width	Minimum Value	Maximum Value
Team	0.00	(Insufficient)	0.00	0.00
Walk Ins	0.00	(Insufficient)	0.00	0.00
Wait Time	Average	Half Width	Minimum Value	Maximum Value
Team	6.7420	(Insufficient)	0.00	32.1345
Walk Ins	0.00	(Insufficient)	0.00	0.00
Transfer Time	Average	Half Width	Minimum Value	Maximum Value
Team	0.00	(Insufficient)	0.00	0.00
Walk Ins	0.00	(Insufficient)	0.00	0.00
Other Time	Average	Half Width	Minimum Value	Maximum Value
Team	3.2191	(Insufficient)	0.00	8.6903
Walk Ins	5.5209	(Insufficient)	3.3880	7.5909
Total Time	Average	Half Width	Minimum Value	Maximum Value
Team	10.2620	(Insufficient)	0.00	40.1436
Walk Ins	6.8009	(Insufficient)	3.3880	9.4409

Queue

Time

Waiting Time	Average	Half Width	Minimum Value	Maximum Value
Seize Barista 1.Queue	7.6292	(Insufficient)	0.00	16.5614
Seize Barista 2.Queue	5.9162	(Insufficient)	0.00	21.2524
Seize Barista 3.Queue	0.00	(Insufficient)	0.00	0.00
Seize Boba 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Boba 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Cashier.Queue	3.1978	(Insufficient)	0.00	22.2537
Seize Cook.Queue	0.1975	(Insufficient)	0.00	2.5600
Seize Milk 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 3.Queue	0.00	(Insufficient)	0.00	0.00
Seize Rice 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Rice 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 3.Queue	0.00	(Insufficient)	0.00	0.00

Other

Number Waiting	Average	Half Width	Minimum Value	Maximum Value
Seize Barista 1.Queue	23.8730	(Insufficient)	0.00	46.0000
Seize Barista 2.Queue	12.2857	(Insufficient)	0.00	30.0000
Seize Barista 3.Queue	5.4840	(Insufficient)	0.00	8.0000
Seize Boba 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Boba 2.Queue	1.6625	(Insufficient)	0.00	2.0000
Seize Cashier.Queue	1.2578	(Insufficient)	0.00	21.0000
Seize Cook.Queue	2.0634	(Insufficient)	0.00	9.0000
Seize Milk 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Milk 3.Queue	0.00	(Insufficient)	0.00	0.00
Seize Rice 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Rice 2.Queue	1.4105	(Insufficient)	0.00	3.0000
Seize Syrup 1.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 2.Queue	0.00	(Insufficient)	0.00	0.00
Seize Syrup 3.Queue	0.00	(Insufficient)	0.00	0.00

Resource

Usage

Instantaneous Utilization	Average	Half Width	Minimum Value	Maximum Value
Bansta	0.9673	(Insufficient)	0.00	1.0000
Boba	0.9185	(Insufficient)	0.00	1.0000
Cashier	0.3885	(Insufficient)	0.00	1.0000
Chicken Karaage	0.00	(Insufficient)	0.00	0.00
Cook	0.5983	(Insufficient)	0.00	1.0000
GyuDon Toppings	0.00	(Insufficient)	0.00	0.00
Milk	0.5499	(Insufficient)	0.00	0.6000
Onigiri filling	0.00	(Insufficient)	0.00	0.00
Rice	0.7722	(Insufficient)	0.00	0.9500
Syrup	0.4400	(Insufficient)	0.00	0.4800
Tako	0.00	(Insufficient)	0.00	0.00

Number Busy	Average	Half Width	Minimum Value	Maximum Value
Bansta	2.9019	(Insufficient)	0.00	3.0000
Boba	18.3706	(Insufficient)	0.00	20.0000
Cashier	0.7431	(Insufficient)	0.00	2.0000
Chicken Karaage	0.00	(Insufficient)	0.00	0.00
Cook	1.6545	(Insufficient)	0.00	3.0000
GyuDon Toppings	0.00	(Insufficient)	0.00	0.00
Milk	21.9976	(Insufficient)	0.00	24.0000
Onigiri filling	0.00	(Insufficient)	0.00	0.00
Rice	15.4433	(Insufficient)	0.00	19.0000
Syrup	21.9976	(Insufficient)	0.00	24.0000
Tako	0.00	(Insufficient)	0.00	0.00

User Specified

Counter

Count	Value
Balking Count	18.0000
Fixed Drink	2.0000
Fruit Tea Count	9.0000
Milk Tea Count	13.0000
Smoothie Count	1.0000
Unfixable Drink	0.00

